

- If the actual number of degrees of freedom is unknown or difficult to compute, it is possible to take the simple k_{GUM} standardized value proposed by the GUM, i.e. $k_{GUM} = 2$ for a coverage probability of 95% or $k_{GUM} = 3$ for a coverage probability of 99%.

When N_E number of effective measurements is higher than 10, the value of k_{IT} is remarkably close to 2, and the values of $U(Z)$ obtained by both solutions are then comparable. Therefore, in all cases where the number of degrees of freedom N_E less than 10, the first solution is advocated. The direct Excel application of these equations is visible on lines 42–45 of the Resource H in Section 5.3.1. The calculation of the relative uncertainty $UR\%$ requires a few comments. By referring to Eq. (6.4), the equation should be:

$$UR\%(Z_m) = \frac{U(Z_m)}{Z_m} \times 100$$

where Z_m is the grand mean of the inverse-predicted concentrations. Unless the analytical method is perfect, it is a biased estimate of X . Therefore, in this context, we recommend using the following formula where X is the assigned value of the corresponding validation material:

$$UR\% = \frac{U(Z_m)}{X} \times 100$$

Because Z_m is biased, it means that the bias would be included in the relative uncertainty and modify the estimation; a positive bias would diminish $UR\%$ and a negative bias would increase it. This point is an important difference between the total analytical error (TAE) concept and the MU, as underlined in Section 6.10 and illustrated in Figure 6.6. Therefore, in the following, the relative uncertainty is noted $UR\%$ to indicate it was calculated from the MAP.

A rapid example from THEOPHYLLINE may illustrate the consequences of using Z_m instead of X . For the first level where $X = 0.05$, the recovered grand mean $\bar{Z} = Z_m = 0.0587$; consequently, the relative bias of about +17%. If $UR\%$ is calculated with Z_m , it gives 39.8% but 46.8% with X .

Finally, it can be asserted that the relative uncertainty better characterizes X . A short remark on this point is available in Section 3.2.2 and a more extended discussion in [21]. To illustrate the diverse calculations presented above, the THEOPHYLLINE dataset is used to give estimates for each validation material and compare the two proposed solutions for the coverage factor. Results are gathered in Table 7.3.

From this table, several conclusions can be drawn:

- For these examples, the GUM standardized coverage interval is always smaller than the coverage interval obtained by the exact method because the number of effective measurements is less than 10, except in one case; the standardized factor k_{GUM} therefore tends to underestimate the expanded uncertainty.
- When the number of effective measurements tends towards the optimum (i.e. 11 in this example), the coverage factor k_{IT} is close to k_{GUM} . The two intervals are almost identical because N_E is optimal.

Table 7.3 THEOPHYLLINE – estimates of expanded uncertainty using two methods of calculating the coverage factor.

Concentration ($\mu\text{g/l}$)	X	0.05	0.10	0.50	1.00	2.50	10.00
Grand mean	$\bar{\bar{Z}}$	0.0587	0.1115	0.5196	1.0013	2.5164	10.352
Standard uncertainty	$u(Z)$	0.0117	0.0130	0.0350	0.0862	0.2749	0.5093
Exact coverage factor							
Effective measures	N_E	7.01	9.59	7.02	5.69	10.91	9.22
Probability β -ETI	β	95%	95%	95%	95%	95%	95%
Coverage factor	k_{IT}	2.36	2.24	2.36	2.49	2.20	2.25
Expanded uncertainty	$U(Z)$	0.028	0.029	0.083	0.214	0.606	1.148
Relative uncertainty	$UR\%$	55.3%	29.2%	16.6%	21.4%	24.2%	11.5%
Coverage interval		0.031	0.082	0.437	0.787	1.91	9.20
		0.086	0.141	0.602	1.216	3.12	11.50
GUM Standardized coverage factor							
Coverage factor	k_{GUM}	2	2	2	2	2	2
Expanded uncertainty	$U(Z)$	0.023	0.026	0.070	0.172	0.550	1.019
Relative uncertainty	$UR\%$	46.8%	26.0%	14.0%	17.2%	22.0%	10.2%
Coverage interval		0.035	0.085	0.450	0.829	1.97	9.33
		0.082	0.138	0.590	1.174	3.07	11.37

Figure 7.3 provides a graphical representation of the coverage intervals, which ultimately appear close. An interesting assumption is that the standard deviation of the β -ETI, s_{IT} , estimates most, if not all, of the combined standard uncertainty since it is limited to the analytical part of MU. The sources of uncertainty excluded by this approach are related to the sampling part and/or the validation material MU. The topic of sampling is discussed in Section 8.3.

Considering the reference values of the validation materials, other sources of uncertainty can be integrated into the model, especially in the case where certified reference materials are used. Sometimes the reference values are obtained through another method of analysis. This is quite common when it is possible to use a reference method, sometimes named *gold standard* method, as in the American literature.

7.3 Use Control Charts Data

7.3.1 Principles of the Shewhart Control Chart

The emergence of control charts in industry dates back to 1924 when W. A. Shewhart (1891–1967) introduced the first version of this quality control (QC) tool in the Bell Telephone Company, where he was employed. Since then, various

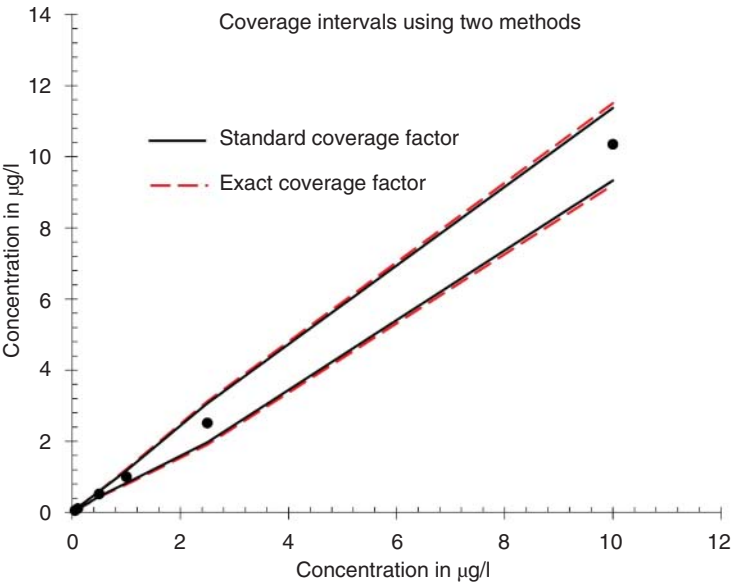


Figure 7.3 THEOPHYLLINE – 95% coverage intervals. Dotted lines: tolerance interval method. Solid lines: standardized GUM.

control charts have been designed, each one adapted to a particular production context, as illustrated by the following (nonexhaustive) list, which consists of the ISO 7870 series of standards relating to control charts:

Ref.	Year	Title
7870-1	2020	General guidelines
7870-2	2013	Shewhart control charts (replaces ISO 8258:1991)
7870-3	2012	Control cards for acceptance
7870-4	2011	Process adjustment control charts
7870-5	2014	Specific control cards
7870-6	2016	EWMA control charts (exponentially weighted moving average)
7870-7	2020	Multivariate control charts
7870-8	2017	Chart techniques for small series and small batches
7870-9	Project	Autocorrelated process control charts

It is not our purpose to develop a complete theory about these various approaches, since it is available in many specialized books [22]. Our purpose is to show how to take advantage, *a posteriori*, of the data accumulated using a control chart to compile an estimate of MU. But several compilation methods are possible according to the type of chart.

This chapter is deliberately limited to the type most traditionally used in laboratories, namely the Shewhart control chart on the mean. Its principle is to

take a “sample of constant size at regular intervals.” Unfortunately, the term *sample* remains highly ambiguous: for an analyst, a sample is a single object that must be analyzed; for a statistician, it is a set of entities forming a sample whose main characteristic is its size.

In other words, for the analyst, the principle of the Shewhart control chart implies, at a given interval of time, taking several test portions of one reference material and making replicates. In laboratories, such reference material is often referred to as QC sample. Let us note J with $2 \leq j \leq J$ the number of replicates. This notation corresponds to the one used for the number of replicates per series of the accuracy profile experimental design.

In an analytical laboratory, the most conventional organization of a control chart consists in inserting at regular intervals QCs among routine samples forming a series. QCs are analyzed by the method, and the measurement values are graphically compared to a target value, denoted T (for Target). Knowledge of the target value has, therefore, a major importance. As it is a question of trueness, Section 4.4 explains how the assigned *true* value T of the QC reference material can be established and the warning and control limits used to verify that any new QC measure is not “out of specification” (OOS) and can be considered as acceptable. Different rules are proposed to make this decision.

When at least one new measurement value deviates from one of these limits, it is concluded that the analytical process presents unexpected conduct. In this case, various corrective actions must be taken before using the method for the following routine analysis. In the everyday vocabulary of the control chart, it is alleged that the process must remain under “statistical process control” (SPC) [22].

In the industrial context of standardized product manufacturing, such as a drug or a food, the target value T can be an expected technical specification, such as the nominal contents of active ingredients, the packaging weight, and so on. For contract laboratories with a wide variety of samples, an internal reference material (IRM or ERM) can be used to set up a control chart because certified reference materials are unavailable or too expensive.

The reference value T , as well as the uncertainty, must have been predetermined in the same context of routine laboratory work. It is also popular to have simultaneously several control charts, and Figure 4.5 illustrates such a framework.

The dataset chosen for illustration is called ALBUMIN. It is available in an official document printed by the European Directorate for the Quality of Medicines and Healthcare (EDQM) devoted to the European medical and clinical laboratory network (OMCL Network of the Council of Europe [GEON]) [11]. It concerns the QC of purified commercial albumin solutions. Albumin solution mainly contains a unique monomer with a molecular weight of 66 kilodaltons (kDa), in addition, a variable percentage of polymers may be present, which are by-products occurring during fractionation, heating, and storage of the solution. Total polymer content (expressed in g/100 ml) is a quality criterion of this impurity, and a low measured value allows for good stability of the solution.

The high-performance liquid chromatography (HPLC) method for checking the total polymer content is based on exclusion chromatography (SEC) [23]. The

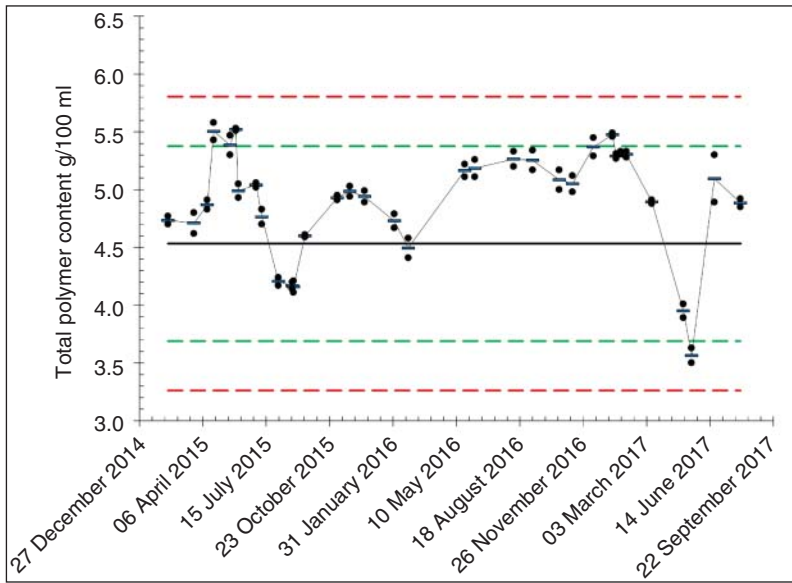


Figure 7.4 ALBUMIN – control chart and QC. Warning limits are in green and control limits are in red.

impurity content of the reference material was determined exactly before starting. QC measures are denoted Z_{ij} where $1 \leq i \leq I$ is the number of times, generally the number of days and/or each use of the method. From the number of replicates J for each QC, three statistics are obtained: the mean, the range, and the standard deviation (equal to the range if $J = 2$). This leads to three types of charts. The most frequently encountered type, the Shewhart control chart, deals with the mean with J small but greater than 1. If $J = 1$, another statistical processing must be applied, which will not be explained here.

QC measurements are plotted on a graph, i.e. Figure 7.4, where the interval between two QC is reported on the horizontal axis. On the vertical axis, the central line represents the assigned target value T of the reference material. When using an IRM, this is usually the grand mean $\bar{\bar{Z}}$ of a set of replicates obtained during a preliminary study with the method to be controlled or a comparable alternative method. On the same axis, the following four limits, going from the top to the bottom, are added, having the general formulas, where T is the target value:

Upper control limit

$$UCL = T + k_1 \frac{\hat{\sigma}}{\sqrt{J}} \quad (7.11)$$

Upper warning limit

$$UWL = T + k_2 \frac{\hat{\sigma}}{\sqrt{J}} \quad (7.12)$$